

Quad Coupler

This is a personal project I did to learn more about working with ADS.

Prompt: Design and simulate a 5.8 GHz branch-line hybrid quadrature coupler in ADS. Target equal power distribution (3dB split) between output and coupled ports with a 90 degree phase separation, maximum isolation, and 50 ohm system impedance.

Why: The hybrid coupler is one of the foundational passive components in RF design. It is used in amplifiers, IQ modulators, mixers, antenna feed networks, splitters and power combiners. At 5.8 GHz you are designing in the upper ISM band used by WiFi, drone control links, and FPV video systems.

Step 1: Understanding the Quad Coupler

The branch line hybrid coupler is a 4 port passive network. It is built from 4 quarter wavelength transmission line sections forming the arms. The “through” arms have characteristic impedance of 50 ohm while the other two arms (called “shunt” arms) have characteristic impedance of 35.35 ohms.

When a signal enters Port 1, the quarter wave sections introduce specific phase delays that cause signals to add constructively at the output port (Port 2) and coupled port (Port 3), splitting power equally at -3dB each. The output and coupled ports carry signals that are 90 degrees out of phase with each other (hence quadrature) this is what makes it valuable in IQ signal processing and balanced amplifier topologies.

The phase delays add destructively at the isolation port (Port 4). All four ports are matched to 50 ohms at the design frequency and performance degrades as frequency drifts.

Step 2: Transmission Lines

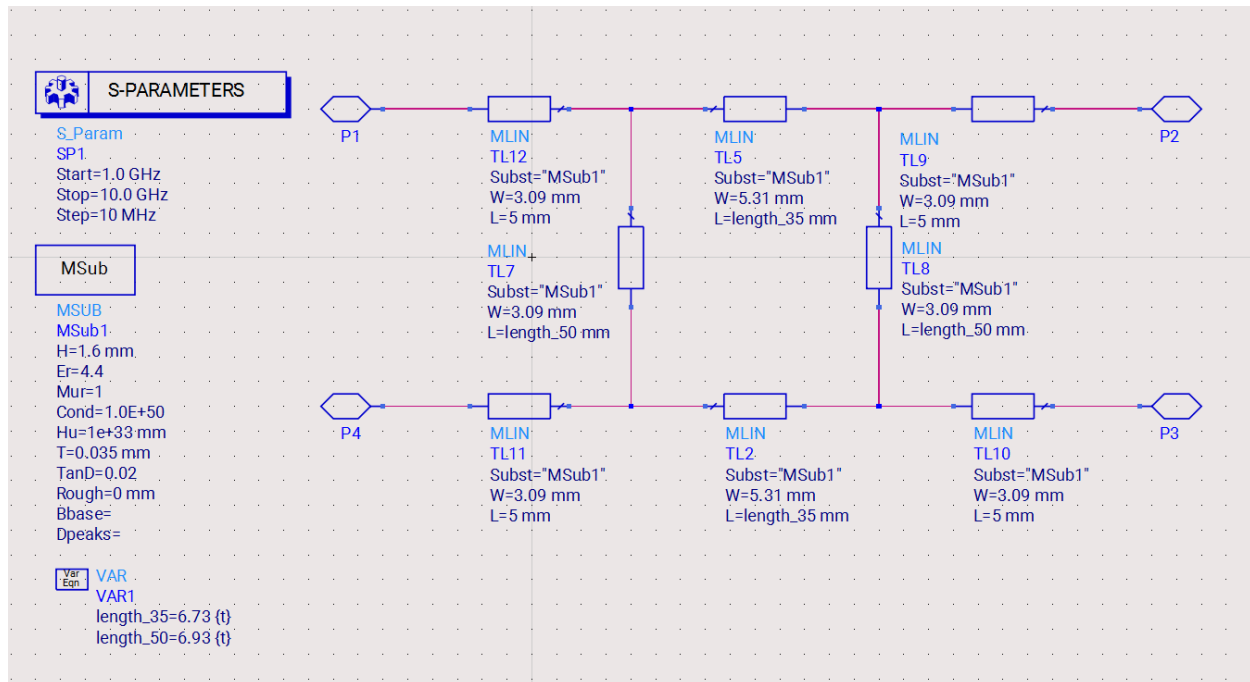
Using Linecalc, calculate the length of 50 ohm lines using FR4 ($\epsilon_r = 4.4$, $H = 1.6\text{mm}$, $T = 0.035\text{mm}$, $\tan\delta = 0.02$). With $Z_0 = 50$ and $E_{\text{eff}} = 90$, Linecalc returns width = 3.09mm and length = 6.93 mm.

Repeating the calculation at 35.35 ohm lines returns width = 5.31mm and length = 6.73mm.

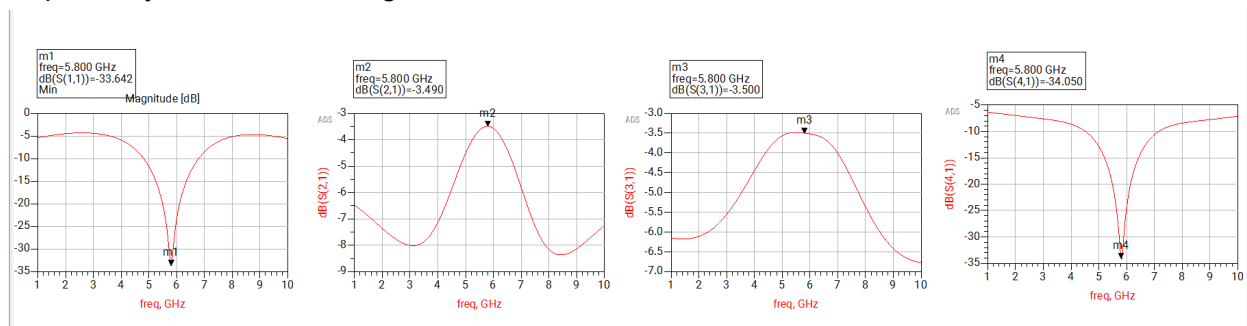
Step 3: Drawing the Schematic

Following the classic quadrature coupler pattern, create the schematic below.

Adding M-Tees to the schematic is more accurate, but interestingly it degrades performance in both the schematic simulation and in Momentum.

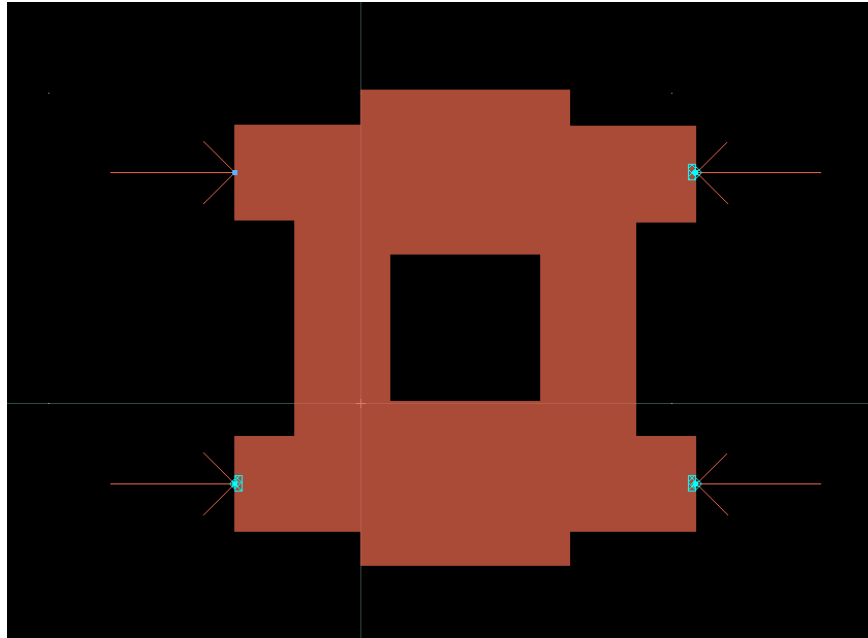


Running the simulation produces the following. S11 shows a dip below -35 dB at exactly 5.8 GHz, while power is divided almost evenly between ports 2 and 3 at -3.49 and -3.5 dB respectively. Isolation is also good, with S41 at -34 dB.

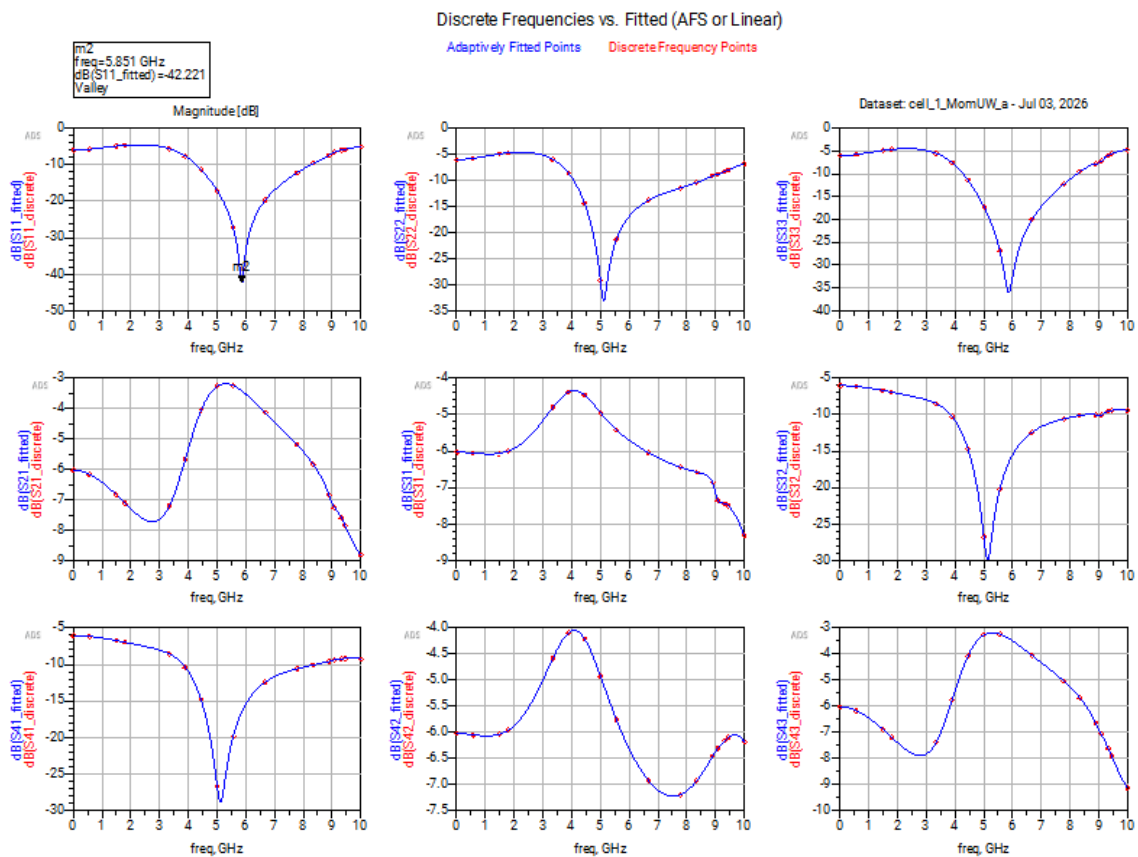


Step 5: Momentum Simulation

After running the initial S-parameter simulation, we can run a more accurate simulation by converting the design to a layout and running a Momentum simulation, which models the EM fields around the copper in the design.

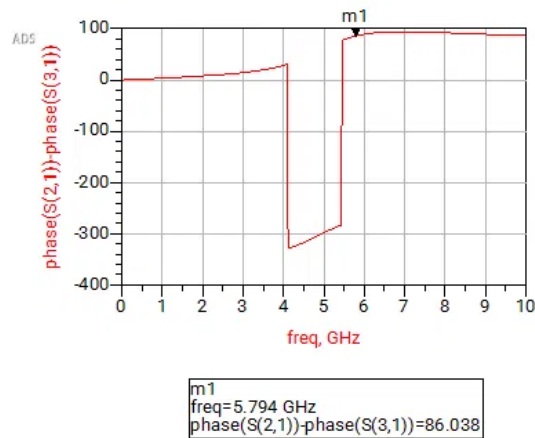


Running the Momentum simulation gives the following S-parameters:



Here, S11 has shifted to -42 dB at 5.85GHz. The power is almost evenly divided, showing -3.5 dB and -4.3dB at ports 2 and 3. The most concerning shift is the isolation, which shows a dip at

5.1 GHz instead of the target frequency. However, the phase is nearly correct, showing 86 degrees difference between S21 and S31 at just below 5.8 GHz.



Step 6: Summary

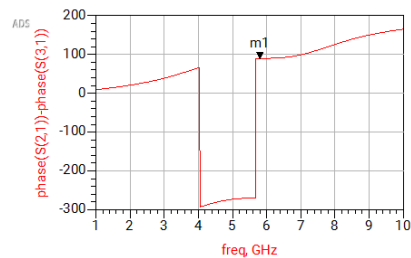
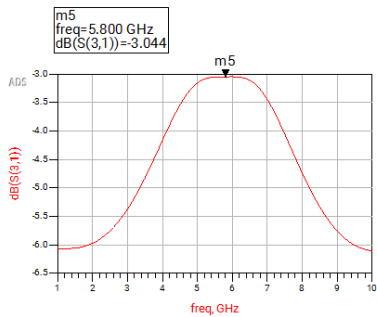
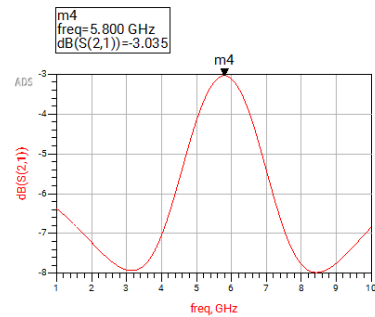
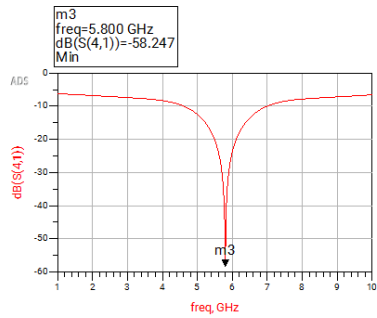
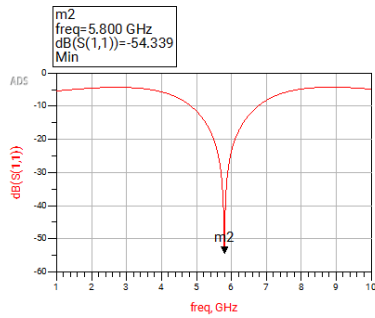
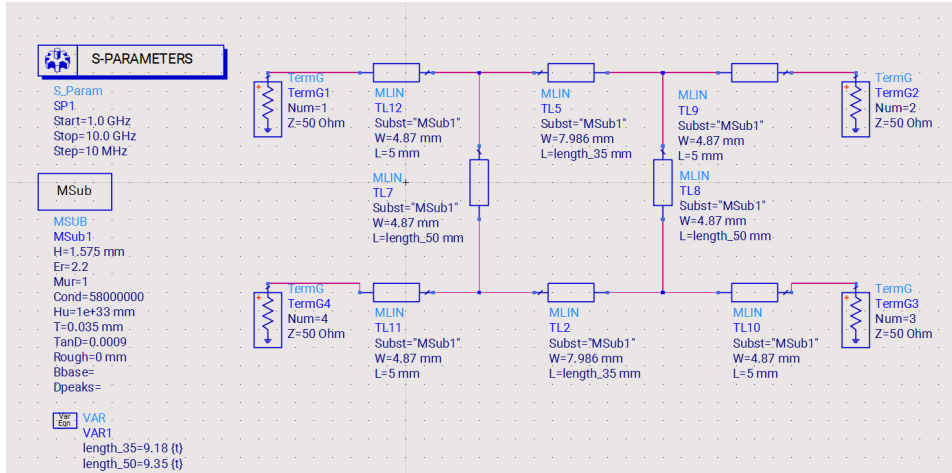
Parameter	Target	Schematic	Momentum
Center frequency	5.8 GHz	~5.8 GHz	5.851 GHz
S11 return loss	deep null	-33 dB	-42 dB
S21	-3 dB	-3.169 dB	-3.5 dB
S31	-3 dB	-4.137 dB	-4.53 dB
S41 isolation	deep null	-34 dB	good
Phase difference	90°	—	86°

Step 7: Differences and Improvements

The differences between the schematic and momentum simulation should be expected; momentum accounts for coupling and line losses which are not represented in the ideal simulation. This accounts for the -4 dB output at the coupled port because some power is lost as heat in the lines. The isolation port also sees some power, which should be expected in a non-ideal component.

The resonant frequency of the coupler is slightly higher than intended - this is acceptable because real-world components usually have some losses which are not modeled in the simulation. The frequency can be designed slightly higher in anticipation of these losses.

Lastly, this coupler was designed on FR4 substrate, which is inexpensive, but is also lossy at high frequencies. A better substrate choice would have been RT duroid 5880. After recalculating line widths, the new substrate gives following S-parameter results:



m1
freq=5.800 GHz
phase(S(2,1))-phase(S(3,1))=89.995